

Course Title: **Big Data and Data Analysis**

Course Code: ICT. Ed 484

Level: Bachelor

Semester: Eighth

Nature of course: Theoretical + Practical

Credit Hour: 3 (2 + 1)

Teaching Hour: 64 (32+32)

1. Course Description

This course provides an extensive overview of big data topics, tools, and methodologies. Students will examine the features and uses of big data, acquire foundational knowledge of Hadoop and its ecosystem, and obtain practical experience in data visualization via Power BI. The course includes data pretreatment methodologies and data analysis with Python modules such as NumPy, Pandas, and PySpark. Practical exercises will consolidate theoretical understanding and generate real-world experience in large data analysis and visualization.

2. General Objectives

The general objectives of this course are as follows:

- Explain the definition and attributes of big data, including volume, velocity, diversity, truth, and value.
- Examine the utilization of big data within the educational framework by comparing big data with conventional data methodologies.
- Discuss the Hadoop ecosystem, encompassing HDFS, MapReduce, YARN, and Hadoop applications such as Hive, Spark, Sqoop, and Pig.
- Integrate Power BI with diverse data sources to develop dynamic dashboards and reports.
- Prepare and modify data for extensive data analysis, encompassing the management of absent and inconsistent data.
- Conduct data analysis utilizing Python libraries (NumPy, Pandas, PySpark) and show data findings.

3. Course Outlines:

Specific Objectives	Contents	Hours (Th+Pr)
• Define the key characteristics of big data. • Analyze the applications of big data in the education system. • Compare big data with traditional data to highlight their differences and advantages.	1. Concept of Big Data 1.1 Definition and Characteristics (Volume, Velocity, Variety, Veracity, Value) 1.2 Applications in Education System 1.3 Big Data vs. Traditional Data 1.4 Recent Trends in Big Data and Practices in Education system	4
• Describe the components of the Hadoop ecosystem.	2. Basic Concepts of Hadoop 2.1 Hadoop Ecosystem Overview 2.2 Hadoop Distributed File System (HDFS) 2.3 MapReduce Framework	8+8

- Explain the functionality of the Hadoop Distributed File System (HDFS). - Implement a simple MapReduce program. - Explore resource management using Hadoop YARN. - Utilize Hadoop applications such as Hive, Spark, Sqoop, and Pig for data processing tasks.	2.4 Hadoop YARN 2.5 Concept of Hadoop Applications: Hive, Spark, Sqoop, Pig <u>Practical Works:</u> - Install and configure Hadoop on a local machine. - Perform basic HDFS operations (upload, download, delete files). - Write and execute a simple MapReduce program. - Explore Hadoop YARN resource management.	
- Identify the components and features of Power BI. - Connect Power BI to various data sources. - Transform and model data within Power BI. - Create interactive visualizations and dashboards. - Collaborate and share reports using Power BI.	3. Data Visualization with Power BI 3.1 Power BI Overview and Components 3.2 Connecting to Data Sources 3.3 Data Transformation and Modeling 3.4 Creating Visualizations 3.5 Sharing and Collaborating with Power BI <u>Practical Works:</u> - Connect Power BI to various data sources. - Transform and model data in Power BI. - Create interactive dashboards and reports. - Share reports and collaborate with team members.	6+6
- Define the importance of data cleaning in big data analytics. - Recognize the impact of missing and inconsistent data on analysis results. - Explain the ETL process and its role in data integration. - Describe principles of data integration and aggregation. - Define the need for data reduction in big data analytics.	4. Data Preprocessing for Big Data 4.1 Data Cleaning and Transformation 4.2 Handling Missing and Inconsistent Data 4.3 Introduction to ETL (Extract, Transform, Load) Processes 4.4 Concept of Data Integration and Aggregation 4.5 Data Reduction and Feature Selection	4+ 2
- Define the basic concepts and importance of data analysis and visualization in various fields. - Setup the key Python libraries (NumPy, Pandas,	5. Data Analysis Using Python Libraries 5.1 Introduction to Data Analysis and Visualization 5.2 Concept of python library and bigdata analysis environment 5.3 NumPy and Statistical functions	8+8

Matplotlib) and describe their roles in big data analysis. • Apply NumPy to create arrays and perform statistical functions such as mean, median, and standard deviation. • Clean and preprocess data using Pandas, including handling missing values and data aggregation. • Create and customize various types of plots (line, bar, scatter) using Matplotlib to visualize data effectively. • Load and explore large datasets from various sources (e.g., CSV, JSON) to gain insights and prepare for analysis.	5.4 Pandas for Data Handling 5.5 Matplotlib for Data Visualization 5.6 Data Ingestion and Exploration in python <u>Practical Works:</u> • Installing and configuring Python and essential libraries (e.g., NumPy, Pandas, PySpark). • Create a NumPy array and perform basic operations (addition, subtraction, mean, standard deviation). • Load a dataset into a Pandas DataFrame, clean the data, and handle missing values. • Perform data aggregation and grouping on a Pandas DataFrame. • Create a line plot and a bar chart using Matplotlib with customized labels and titles. • Loading large datasets from various sources (e.g., CSV, JSON, databases).	
• Define the basic concepts of big data and the role of PySpark in big data analysis. • Install and configure PySpark to set up the big data analysis environment. • Create and manipulate PySpark DataFrames for data analysis. • Perform large-scale data analysis using PySpark to gain insights. • Query data using Spark SQL to extract meaningful information from PySpark DataFrames. • Create interactive visualizations and generate reports to summarize data insights.	6. Big Data Analysis Using Python Libraries 6.1 Introduction to Big Data and PySpark 6.2 Installing and configuring PySpark 6.3 PySpark DataFrames 6.4 Big Data Analysis with PySpark 6.5 Using SQL (DBMS tools) for querying data 6.6 Data Visualization and Reporting 6.7 Concept of Spark SQL and its application <u>Practical Works:</u> • Set up a PySpark environment and create a simple PySpark application. • Load a large dataset into a PySpark DataFrame and perform basic operations (filtering, grouping). • Handle missing data in a PySpark DataFrame and perform data cleaning. • Use Spark SQL to query a PySpark DataFrame and analyze the results. • Create interactive visualizations and dashboards using Pandas with Matplotlib/Seaborn and PySpark with Plotly. Generate a report summarizing the insights from the visualizations.	6+8

4. Instructional Techniques

The instructional techniques for this course are divided into two groups. First group consists of general instructional techniques applicable to most of the units. The second group consists of specific instructional techniques applicable to specific units.

4.1 General Techniques

- Providing the reading materials to the students to familiarize the units.
- Lecture, question-answer, discussion, brainstorming, practical, hands-on labs, project based learning and flipped classrooms.

4.2 Specific Instructional Techniques

Unit	Activity and instructional techniques	Teaching Hours
1	• Use real-world examples to explain each characteristic of big data (Volume, Velocity, Variety, Veracity, Value) • Discuss case studies where big data has been used to improve educational outcomes. • Highlight the differences in data processing and analysis techniques. • Present current research and developments in big data practices	64
2	• Provide an overview of HDFS, MapReduce, YARN, and Hadoop applications. • Demonstrate basic HDFS operations (upload, download, delete files). • Write and execute a simple MapReduce program. • Discuss the role of YARN in managing cluster resources. • Provide examples of how each application is used in big data processing.	
3	• Introduce the Power BI interface and its main features. • Demonstrate how to connect Power BI to different data sources (e.g., Excel, databases). • Show how to clean and prepare data for analysis in Power BI. • Guide students through creating various visualizations (e.g., charts, graphs). • Explain how to publish and share Power BI reports.	
4	• Teach techniques for handling missing and inconsistent data. • Provide strategies for dealing with incomplete data. • Explain the steps involved in extracting, transforming, and loading data. • Discuss methods for combining and summarizing data. • Introduce techniques for data reduction and feature selection.	
5	• Provide an overview of data analysis techniques and visualization tools. • Discuss the roles of NumPy, Pandas, and PySpark in data analysis.	

	• Demonstrate how to perform basic operations with NumPy arrays. • Show how to load, clean, and manipulate data in Pandas DataFrames. • Guide students through creating and customizing plots. • Teach techniques for loading and exploring large datasets.	
6	• Introduce PySpark and its capabilities. • Provide step-by-step instructions for setting up PySpark. • Demonstrate basic operations with PySpark DataFrames • Guide students through analyzing large datasets with PySpark. • Show how to write and execute SQL queries in PySpark. • Teach students how to create interactive visualizations and summarize insights in reports.	

5. Evaluation (Internal Assessment and External Assessment):

Nature of course	Internal Assessment	External Practical Exam/Viva	Semester Examination	Total Marks
Theory	40%	20%	40%	100%

Note: Students must pass separately in internal assessment, external practical exam / viva and or semester examination.

5.1 Evaluation for Part I (Theory): Internal Evaluation 40%

Internal evaluation will be conducted by course teacher based on following activities:

1) Attendance	5 points
2) Participation in learning activities	5 points
3) First assessment (written assignment)	10 points
4) Second assessment (Term examination)	10 points
5) Third assessment (Internal Practical Exam/Case Study)	10 points
Total	40 points

5.2 External Evaluation (Final Examination) 40%

Examination Division, office of the Dean, Faculty of Education will conduct final examination at the end of semester.

1) Objective type question (Multiple choice 10questionsx1mark)	10 marks
2) Short answer questions (6 questions x 5 marks)	30 marks
Total marks	40

5.3 Evaluation for part II (practical) 20%

Nature of the course	Semester final examination by External Examiner	Total percent
Practical	100%	100

5.3.1 Practical Examination Evaluation Scheme

- a) External assessment.....100%
 - i) Lab Report/Project Report 20%
 - ii) Laboratory work exam/Case 40%
 - iii) VIVA.....40%

6. Recommended books and reading materials (including relevant published articles in national and international journals)**6.1 Prescribed Text Book****6.2 Reference Materials**

1. Mayer-Schönberger, V., & Cukier, K. (2014). *Big Data: A Revolution That Will Transform How We Live, Work, and Think*. Houghton Mifflin Harcourt.
2. White, T. (2012). *Hadoop: The Definitive Guide*. O'Reilly Media.
3. Karau, H., & Warren, R. (2020). *Learning Spark: Lightning-Fast Data Analytics*. O'Reilly Media.
4. Knight, D. (2022). *Microsoft Power BI Quick Start Guide*. Packt Publishing.
5. Powell, B. (2024). *Mastering Microsoft Power BI*. Packt Publishing.
6. McKinney, W. (2017). *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython*. O'Reilly Media.
7. Grus, J. (2019). *Data Science from Scratch: First Principles with Python*. O'Reilly Media.
8. Holmes, D. E. (2017). *Big Data: A Very Short Introduction*. Oxford University Press.
9. Balusamy, B., Abirami, N. R., Kadry, S., & Gandomi, A. (2021). *Big Data: Concepts, Technology, and Architecture*. Wiley.